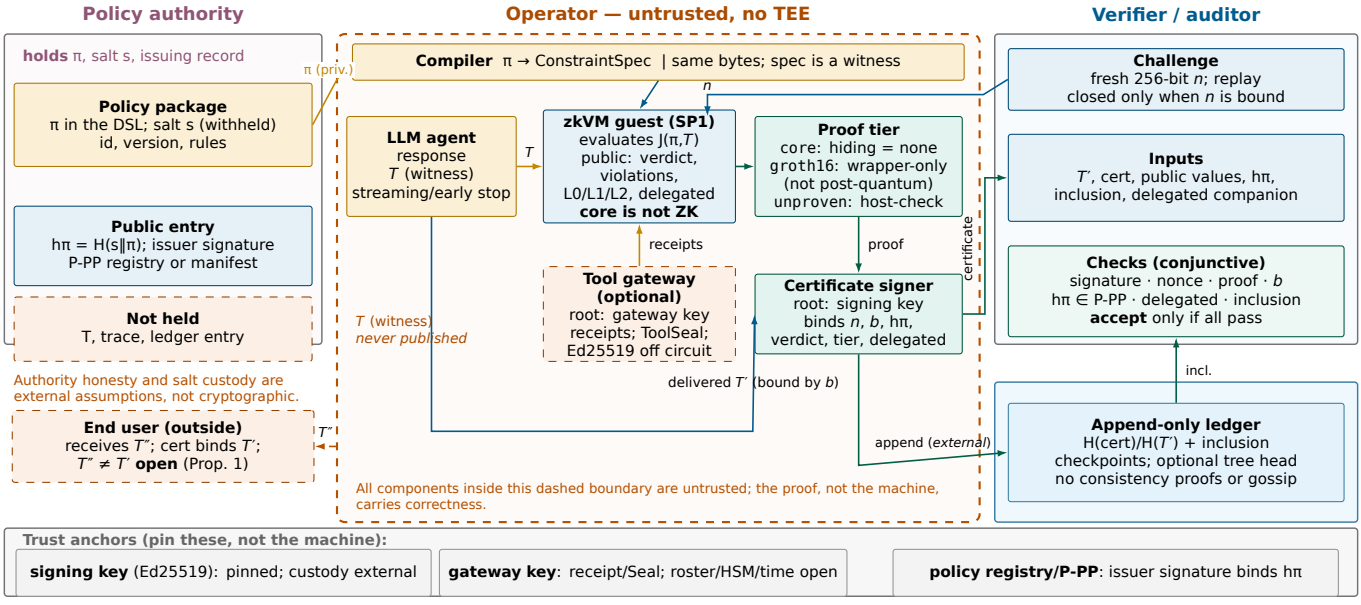
Policy authority
Operator — untrusted, no TEE
Verifier / auditor
holds π, salt s, issuing record
π (priv.)
Policy package
π in the DSL; salt s (withheld)
id, version, rules
Public entry
hπ = H(s‖π); issuer signature
P-PP registry or manifest
Not held
T, trace, ledger entry
Authority honesty and salt custody are external assumptions, not cryptographic.
End user (outside)
receives T″; cert binds T′;
T″ ≠ T′ open (Prop. 1)
T″
Compiler π → ConstraintSpec | same bytes; spec is a witness
n
LLM agent
response
T (witness)
streaming/early stop
T
zkVM guest (SP1)
evaluates J(π,T)
public: verdict,
violations,
L0/L1/L2, delegated
core is not ZK
Proof tier
core: hiding = none
groth16: wrapper-only
(not post-quantum)
unproven: host-check
receipts
proof
Tool gateway
(optional)
root: gateway key
receipts; ToolSeal;
Ed25519 off circuit
Certificate signer
root: signing key
binds n, b, hπ,
verdict, tier, delegated
certificate
T (witness)
never published
delivered T′ (bound by b)
append (external)
Challenge
fresh 256-bit n; replay
closed only when n is bound
Inputs
T′, cert, public values, hπ,
inclusion, delegated companion
Checks (conjunctive)
signature · nonce · proof · b
hπ ∈ P-PP · delegated · inclusion
accept only if all pass
incl.
Append-only ledger
H(cert)/H(T′) + inclusion
checkpoints; optional tree head
no consistency proofs or gossip
All components inside this dashed boundary are untrusted; the proof, not the machine, carries correctness.
Trust anchors (pin these, not the machine):
signing key (Ed25519): pinned; custody external
gateway key: receipt/Seal; roster/HSM/time open
policy registry/P-PP: issuer signature binds hπ
public
witness
signed / anchored
external / open
crypto binding
external assumption
open gap
Design, not measured. No TEE. core leaves the witness open; groth16 hiding is wrapper-only and not post-quantum; L-side (re-execution) is open at every tier.
External: authority salt; gateway key; append-before-delivery; no split view; T″ binding.